



Norm-enhanced prebunking for actively open-minded thinking indirectly improves misinformation discernment and reduces conspiracy beliefs

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ABSTRACT

Recent research has demonstrated that actively open-minded thinking (AOT)—a cognitive thinking style characterized by the active avoidance of myside bias and overconfidence in one's conclusions—is related to lower misinformation susceptibility. Furthermore, logic-based inoculation has proven effective at conferring resistance against misinformation and conspiracy beliefs. Building on these findings, the current article outlines two pre-registered experiments, conducted on [Reddit.com](https://www.reddit.com) (Study 1, $N = 462$) and *Prolific* (Study 2, $N = 464$), wherein participants were either allocated to a control condition or presented with an inoculation message prebunking failure to engage in AOT. We hypothesised that improvements in AOT relative to the control group should indirectly reduce misinformation susceptibility and conspiracy beliefs. Results showed that in both studies, the AOT inoculation significantly improved AOT relative to the control group, and in turn, reduced conspiracy beliefs and improved veracity discernment of real news headlines from fake ones. Furthermore, the intervention also improved sharing decisions (Study 1), and its efficacy was exclusive to improving AOT rather than its individual epistemic components (Study 2). We provide recommendations for future efforts to reduce misinformation susceptibility using logic-based inoculation and discuss the theoretical implications of our findings.

The spread of harmful misinformation presents a threat to science, public health, and democracies worldwide (Ecker et al., 2024; van der Linden & Kyrychenko, 2024). Examples abound. From 5G conspiracy theories leading people to set phone masts on fire (Jolley & Paterson, 2020) to fake news stories about immigrants fuelling racism and national riots (Venkataramakrishnan, 2025) to anti-vaccine misinformation undermining vaccination coverage (Pierri et al., 2022). Perhaps unsurprisingly, the field of psychology has witnessed a sharp rise in research seeking to understand the psychological mechanisms behind why people are susceptible to misinformation and conspiracy theories (Biddlestone, Green, et al., 2025; Sultan et al., 2024) and what interventions might be effective in countering its spread (Ecker et al., 2023; van der Linden, 2022; Roozenbeek & van der Linden, 2024). The APA recently defined misinformation as “any information that is demonstrably false or misleading regardless of its source or intention” (American Psychological Association, 2023, p. 7). The APA report also identified

several promising interventions, including psychological inoculation. Psychological inoculation is a social psychological theory (McGuire, 1964; van der Linden, 2024) and intervention that can be used to reduce misinformation susceptibility (e.g., Lu et al., 2023; van der Linden, 2024). Originally theorised by McGuire (1964), psychological inoculation follows the medical analogy and presents participants with a weakened dose of misinformation and persuasive preemptive refutations of that misinformation akin to how the body produces immunity against disease via vaccination. Many reviews have pointed to its efficacy in inducing resistance to (malicious) persuasion (see Banas & Rains, 2010; Compton et al., 2021; Lu et al., 2023; Traberg et al., 2023; van der Linden, 2023, 2024). Interestingly, Pennycook et al. (2024) recently reported that neither psychological inoculation nor accuracy nudges—brief psychological prompts to consider the accuracy of information before engaging with it—were effective by themselves at improving discernment of manipulative content from non-manipulative

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content. However, when the two interventions were combined (i.e. when inoculation was paired with the additional presentation of accuracy prompts), the intervention helped people discern manipulative content from non-manipulative content more effectively. Despite meta-analytic work indicating that inoculation is effective alone (Huang et al., 2024; Lu et al., 2023), Pennycook et al.' (2024) work raises an interesting question: can the effectiveness of inoculation (or “prebunking”) messages be enhanced—or at least ensured—with a focus on improving cognitive thinking styles that are predictive of misinformation discernment? In this article, we set out to explore whether prebunking the failure to engage in actively open-minded thinking (AOT) can reduce misinformation susceptibility and conspiracy beliefs.

1. Actively open-minded thinking and misinformation susceptibility

An extensive scholarly debate around the psychological predictors that account for misinformation susceptibility has occurred in recent years. On the one hand, proponents of the *lazy, not biased* perspective argue that inaccurate beliefs can be primarily explained by a lack of engagement of the cognitive processes that favour accuracy (i.e., cognitive reflection; see Pennycook & Rand, 2019). On the other hand, proponents of the contemporary motivated reasoning account argue that accuracy goals compete with identity-related goals, wherein the truth is rejected (rather than ignored) when identity-related goals are favoured (see Van Bavel & Pereira, 2018). Roozenbeek, Maertens, et al. (2022) conducted a comprehensive comparison of these two perspectives, showing that partisanship and myside bias (a central component to AOT) accounted for more variance in misinformation susceptibility than cognitive reflection alone (see also Bowes et al., 2023 for a meta-analysis demonstrating that AOT is also a negative predictor of conspiracy beliefs). Consequently, the authors argued for a more integrative account of the cognitive predictors of misinformation susceptibility.

Ultimately, Baron (2019) defines AOT as a cognitive thinking style characterized by the active avoidance of myside bias and overconfidence in one's conclusions. Baron (2019) also argues that AOT—like cognitive reflection—encompasses the motivation to remain accurate in one's appraisal of information. However, unlike cognitive reflection, AOT also contains several other important elements. Most importantly, AOT also details the roles of overconfidence in one's opinion and the overestimation of one's knowledge on a subject. This carries important implications for how individuals respond to contradictory evidence once they have engaged in cognitive reflection. Theoretically, failure to engage in AOT leads to an unwillingness to shift positions when information refuting one's attitude is encountered, referred to as *myside bias* (or low intellectual humility). Additionally, overconfidence in one's knowledge also accounts for the *illusion of explanatory depth*, captured by conviction on a subject until tasked with explaining how and why this conviction was formed.

Failure to engage in these central tenets of AOT reduces the motivation to search for information that would otherwise provide a more informed opinion on a subject. More specifically, failure to engage in AOT is reflected in failure to consider alternative hypotheses, belief overkill (i.e., the interpretation of contradictory evidence as support for one's opinion), and selective attention to supporting evidence. As such, AOT is a multifaceted cognitive thinking strategy that has far-reaching implications for how individuals perceive, accept, and search for information. Finally, Baron (2019) argues that AOT is a socio-cognitive cultural norm that can be socially encouraged (its antithesis demonstrated by norms encouraging blind loyalty), rather than a cold cognitive mechanism per se. Taken together, these elements suggest that an effective intervention focusing on improving the implementation of AOT is likely to result in a myriad of different cognitive processes that reduce the likelihood of accepting falsehoods. We propose that this approach should work in a similar way to Pennycook et al.' (2024) combination of cognitive reflection and inoculation. Extending on their approach, we

will take a more direct focus on AOT, which captures a more integrative combination of important cognitive components that are important in helping people discern misinformation from reliable information.

2. Types of inoculation: from technique-based to logic-based interventions

There are a number of different ways that psychological inoculation can be administered. Most of the misinformation literature has employed technique-based inoculation, prebunking the strategies that *merchants of doubt* (Oreskes & Conway, 2010) may use to manipulate people. While this can include issue-based inoculation, which focuses on prebunking misinformation about specific claims or topics such as casting doubt on the scientific consensus on climate change (van der Linden et al., 2017), technique-based inoculation generally aims to immunise people against the larger tactics used to manipulate people (Biddlestone, Roozenbeek, et al., 2025; Lewandowsky & van der Linden, 2021; van der Linden, 2024). For example, the *Bad News* game places the player in the position of someone spreading weakened doses of disinformation with the use of six separate strategies: spreading conspiracy theories, fuelling polarisation, using emotional language, trolling, deflection, and impersonation (e.g., Basol et al., 2020). This approach has proven effective at improving the detection of manipulative social media posts using these same strategies (Traberg & van der Linden, 2022).

One alternative to technique-based inoculation is logic-based inoculation, aiming to confer resistance against the psychological states (e.g., psychological biases, see Biddlestone et al., 2022; anxieties that can hinder our effective performance on tasks, Jackson et al., 2017) that make us vulnerable to manipulation in the first place. Logic-based inoculation may provide broader and longer-lasting psychological resistance against misinformation regardless of the topic or context. Logic-based inoculation has been used in the context of conspiracy theories (e.g., through prebunking the logical fallacies used in conspiracy theories and misinformation; Banas & Miller, 2013; Biddlestone, Roozenbeek, et al., 2025) and is known to confer some degree of cross-protection against other tactics (Roozenbeek, Traberg, & van der Linden, 2022). Extending technique and logic-based inoculation to the broader realm of psychological motives that make people susceptible to misinformation may result in a broader alteration of the psychological states that make us vulnerable to different forms of epistemically unwarranted beliefs. Theoretically, this should provide blanket protection against a whole host of these problematic beliefs. In relation to epistemic motives, Biddlestone et al. (2022) used logic-based inoculation to reduce susceptibility to conjunction errors—a probabilistic cognitive bias captured by the fallacy that two events occurring together can be more likely than two independent events occurring separately (see also Biddlestone, Roozenbeek, et al., 2025 for other logic-based inoculations). Importantly, this approach also indirectly reduced belief in certain conspiracy narratives and improved the detection of both real and fake news headlines relative to a control group. We argue that prebunking failure to engage in AOT falls under the logic-based inoculation framework due to its purpose of explaining a thinking style that makes us vulnerable to manipulation, much like other cognitive biases that are prebunked in the logic-based inoculation literature (e.g., Biddlestone et al., 2022).

2.1. Norm-enhanced inoculation against failure to engage in actively open-minded thinking

The use of interventions to reduce misinformation susceptibility by improving AOT are not new. For example, Mirhoseini et al. (2023) used a computer-based approach to provide feedback on participants' veracity discernment abilities during an item-rating task of real and fake news headlines. The authors reasoned that negative feedback should motivate participants to re-evaluate their beliefs until the feedback

improves. While the re-evaluation of one's beliefs through the appropriate search for evidence is central to AOT (see Baron, 2019), exploring the role of other proposed components of AOT may also be important. For example, the illusion of explanatory depth and intolerance of uncertainty were largely missing from Mirhoseini et al.' (2023) paradigm. Therefore, we sought to implement a more detailed and explicit explanation of AOT to participants with the use of the inoculation framework to attempt to capture the full extent of the components that make up AOT.

Extending Biddlestone et al.' (2022) use of logic-based inoculation against conjunction errors to reduce misinformation susceptibility and conspiracy beliefs, we aimed to use a similar strategy to improve AOT. Considering Baron's (2019) reasoning, inoculation against failure to engage in AOT should improve processes associated with cognitive reflection and intellectual humility, and reduce processes associated with myside bias, overconfidence in one's opinions, and the illusion of explanatory depth. Importantly, Baron (2019) also argues that AOT can be a culturally encouraged norm, implying the ability to improve AOT with the use of norm appeals. Therefore, we also aimed to implement a similar norm-enhanced inoculation message to that administered by Maertens et al. (2020), wherein participants are presented with information communicating consensus agreement with the need to engage in AOT to further enhance the efficacy of our message. Descriptive norm appeals communicating majority consensus are well-documented as effective behavioural interventions (e.g., Goldstein et al., 2008; Halls-worth et al., 2016; Schultz et al., 2018). Thus, we posit that this will be an effective way to communicate the importance of engaging in AOT to participants.

The inoculation strategy employed here is primarily logic-based in its attempt to improve a cognitive thinking style, with an enhancement component of norm appeals to ensure its effectiveness. Finally, in line with research demonstrating the negative relationship between AOT and both misinformation susceptibility (Roozenbeek, Maertens, et al., 2022) and conspiracy beliefs (Bowes et al., 2023), improving AOT should reduce misinformation susceptibility and conspiracy beliefs.

3. Overview of the current experiments

In the current experiments, we employed between-subject designs to determine whether a norm-enhanced inoculation message significantly improves mean AOT scores relative to a control group. Furthermore, in both studies we set out to analyse whether this message would indirectly reduce misinformation susceptibility and conspiracy beliefs relative to the control group, additionally including willingness to share misinformation in Study 1. We argue that an indirect path model is more informative than simply observing direct effects due to the primary intention of our intervention to improve AOT, which should subsequently have knock-on effects for our other outcome variables. Indirect path models also enabled us to determine whether these hypothesised effects remained when controlling for the shared variance with epistemic variables that we either deemed relevant or specifically acting as components of AOT: In both studies we controlled for cognitive reflection, and in Study 2 we additionally controlled for numeracy ability, intellectual humility, intolerance of uncertainty, the illusion of explanatory depth, need for closure, and ideological bias awareness.

Among many other elements, Baron (2019) argues that AOT encompasses reflective thinking. Roozenbeek, Maertens, et al. (2022) demonstrated that AOT uniquely predicted misinformation susceptibility even when controlling for scores on the cognitive reflection task (CRT). Nevertheless, to demonstrate that our intervention improves AOT—and not simply reflective thinking—we included the CRT as a control variable alongside AOT in both studies to account for their shared variance. Meta-analytic evidence also suggests that low AOT is associated with stronger conspiracy beliefs (Bowes et al., 2023; see also Patel et al., 2019; Rizeq et al., 2020). This implies a tendency to resist efforts to alter the conspiracy beliefs that these individuals already hold.

Therefore, we included the conspiracy mentality questionnaire (CMQ; Bruder et al., 2013) as a potential covariate and moderator to determine whether our hypothesised experimental effects are conditional on prior susceptibility to conspiracy theories in Study 1.

4. Study 1

In Study 1, we sought to test our hypotheses that AOT inoculation would improve AOT, and in turn, improve veracity discernment, sharing decisions, and reduce conspiracy beliefs. To measure misinformation susceptibility in various ways (including veracity discernment), we used the Misinformation Susceptibility Test (MIST; Maertens et al., 2023). The MIST is a psychometrically validated instrument designed to measure news veracity discernment. It was developed using a rigorous four-stage process, including automated item generation via GPT-2, expert filtering, and both classical test theory and item response theory for item selection. The final scale includes a balanced set of real and fake headlines and captures multiple facets of misinformation susceptibility, including the ability to detect real news, detect fake news, overall veracity discernment, and judgment biases such as distrust. This multi-factor structure allows for nuanced analysis of how interventions impact different aspects of misinformation processing. Crucially, the MIST addresses key limitations of prior misinformation measures, which often lacked psychometric validation, used unbalanced or unverified items, and failed to distinguish between different cognitive mechanisms underlying susceptibility. Previous ad hoc tests rarely measured latent traits or controlled for judgment biases such as distrust. In contrast, the MIST offers a validated higher-order structure, includes norm tables, and allows for disaggregated interpretation of real/fake news detection and response biases, ensuring greater construct validity and comparability across studies. The MIST has demonstrated robust internal validity, cross-national replicability, and predictive validity in large, preregistered samples, making it a reliable and versatile tool for misinformation research.

In our pre-registration, we hypothesised that the AOT inoculation would significantly increase AOT compared to the control group, and that it would indirectly reduce conspiracy beliefs and intentions to share fake news headlines, and indirectly increase the detection of both real and fake news headlines. We also pre-registered exploratory hypotheses for all other measures. All hypotheses and analyses for Study 1 were fully pre-registered here: https://aspredicted.org/772_BXC.

4.1. Methods

4.1.1. Participants

A priori power analysis using G*Power indicated that to detect an experimental effect size of $d = 0.56$ (equivalent to $r = 0.27$ for the correlation between AOT and ability to detect fake news in a previous unpublished study) with a power of .90 and two experimental groups, a minimum sample size of $N = 136$ was required. To account for potential data exclusions and exploratory analyses, we aimed to use snowball sampling to achieve a minimum sample size of $N = 200$ ($N = 100$ per condition).

The survey was distributed on [Reddit.com](https://www.reddit.com) through various US-based forums. 1215 responses were initially collected, but once we removed participants who had failed the Generic Conspiracist Beliefs Scale (GCBS) attention check ($N = 324$), the experimental manipulation attention check ($N = 186$), that were not US nationals ($N = 175$), or indicated that they had taken part in a similar study by the same researchers previously ($N = 201$), the final sample size was 462 (253 men, 177 women, $M_{age} = 35.39$, $SD_{age} = 11.34$).

Sensitivity power analysis using the "statsmodels" module in Python indicated that a power of .90 was achieved to detect a minimum effect size of $d = 0.35$, and a power of .80 was achieved to detect a minimum effect size of $d = 0.30$. According to Fritz and MacKinnon (2007), we also achieved enough power to detect a mediation effect with small

coefficients on both the *a* and *b* paths of an indirect path model.

4.1.2. Design and procedure

After reading the information sheet and providing their consent, participants were randomly allocated to the treatment or control condition. Afterwards, they were presented with the measures of AOT and CRT in randomised order and were then asked to complete the measures of conspiracy beliefs, misinformation susceptibility, and willingness to share real and fake news in randomised order. Finally, participants filled out measures of conspiracy mentality and other covariates before being fully debriefed.

4.1.3. Materials

In the norm-enhanced inoculation against failure to engage in AOT, participants were presented with the following text:

Please read the following text carefully, considering how arguments and evidence may be received and evaluated.

In a recent survey, we found that 80% of people agreed that they should **actively search for more information** that both **supports and contradicts** their current viewpoints than they currently tend to do **before feeling convinced** on a topic.

For this reason, you should be vigilant of the fact that some online content producers seek to **manipulate their audience** by exploiting the common tendency to feel **confident in the opinions that you already hold**. This overconfidence reduces the likelihood that you will be motivated to **search for relevant information** that would otherwise give you a better understanding of the topic at hand. In psychological research, this prevalent issue is often referred to as **failing to engage in actively open-minded thinking**.

Failing to effectively engage in actively open-minded thinking can be identified through five main pitfalls:

1. **Overconfidence** in your position.
2. Failure to consider **alternative possibilities**.
3. **Conviction** that you understand your position **until asked to explain it**.
4. Only **searching** for and attending to **evidence that supports your position**.
5. **Interpreting** all evidence as **support for your position**, even when it isn't.

As a result, reduced actively open-minded thinking has been linked to many problematic outcomes for society, including **poorer ability to objectively evaluate arguments**, as well as increased susceptibility to **misinformation and conspiracy theories**.

Importantly, actively open-minded thinking does **not require you to be skeptical** of all information and viewpoints you encounter, but rather gives you the tools to more **appropriately evaluate when and who you should trust**. A helpful approach to ensure you are engaging in actively open-minded thinking is to ask yourself whether the content you are consuming provides information **explaining how it avoided** the five pitfalls mentioned above when drawing its conclusions.

So next time you're watching the news or reading information online, remember...**don't believe everything you think!**

The content included in this text aimed to 1) communicate a norm consensus, 2) motivate participants to be defensive against individuals trying to exploit their tendency to fail to engage in AOT, and 3) explain five common signs of failing to engage in AOT. In the control condition, participants were simply told to begin the study when they were ready.

4.1.4. Measures

Actively open-minded thinking ($\alpha = .81$; $M = 4.03$, $SD = 0.61$) was measured with Baron et al.' (2022) 10-item AOT scale (e.g., "Willingness to be convinced by opposing arguments is a sign of good character"), using a response scale from *completely disagree* (1) to *completely agree* (5).

Cognitive reflection ($\alpha = .64$; $M = 2.00$, $SD = 1.08$) was measured using three items that are similar to Frederick's (2005) original CRT. Participants were presented with the following three questions: "If you're running a race and you pass the person in second place, what place are you in?", "A farmer had 15 sheep and all but 8 died. How many are left?", and "Emily's father has three daughters. The first two are named April and May. What is the third daughter's name?". Responses were collected in text format and summed based on whether they were correct (1) or incorrect (0).

Misinformation susceptibility was measured with Maertens et al.' (2023) eight-item misinformation susceptibility test (MIST-8), providing distinct measures of real news detection ($M = 3.08$, $SD = 1.12$), fake news detection ($M = 3.08$, $SD = 1.04$), veracity discernment ($M = 6.16$, $SD = 1.65$), and distrust ($M = 4.01$, $SD = 1.40$) depending on the way scores are calculated. Respondents were presented with four real (e.g., "Attitudes Toward EU Are Largely Positive, Both Within Europe and Outside It") and four fake (e.g., "Certain Vaccines Are Loaded with Dangerous Chemicals and Toxins") news headlines and were asked to decide using a binary scale whether each headline was *fake* or *real* (see also Roozenbeek, Maertens, et al., 2022, for further details). Correct detection of real and fake news headlines was summed through these correct responses using the binary scale. Veracity discernment was calculated by summing the correct detection of both real and fake news headlines. Distrust was calculated by summing the total number of times participants selected *fake*, regardless of whether the headlines were real or fake. Analysis of Skewness and Kurtosis indicated that the data for the detection of both real, Skewness = -1.17 , and fake news headlines, Skewness = -1.21 , were negatively skewed.

Willingness to share real ($\alpha = .86$; $M = 2.74$, $SD = 1.54$) and **fake news headlines** ($\alpha = .87$; $M = 2.09$, $SD = 1.37$) was measured by presenting participants with fake tweets detailing the same real and fake news headlines presented in the MIST-8 (see Supplement, p. 11). A measure of sharing decisions was calculated by subtracting intentions to share fake news headlines from intentions to share real news headlines ($M = 0.65$, $SD = 1.25$). Participants were asked to indicate how likely they would be to share each post using a response scale from *not at all likely* (1) to *extremely likely* (7).

Conspiracy beliefs were measured with Brotherton et al.' (2013) 15-item GCBS, using a response scale from *definitely not true* (1) to *definitely true* (5). The GCBS includes sub-scales measuring conspiracy beliefs regarding government malfeasance (e.g., "The government is involved in the murder of innocent citizens and/or well-known public figures, and keeps this a secret"; $\alpha = .86$; $M = 2.66$, $SD = 1.14$), malevolent global conspiracies (e.g., "The power held by heads of state is second to that of small unknown groups who really control world politics"; $\alpha = .87$; $M = 2.16$, $SD = 1.06$), extra-terrestrial cover-ups (e.g., "Secret organizations communicate with extraterrestrials, but keep this fact from the public"; $\alpha = .87$; $M = 2.07$, $SD = 1.02$), personal wellbeing (e.g., "Experiments involving new drugs or technologies are routinely carried out on the public without their knowledge or consent"; $\alpha = .81$; $M = 2.11$, $SD = 1.00$), and the control of information (e.g., "A lot of important information is deliberately concealed from the public out of self-interest"; $\alpha = .68$; $M = 2.93$, $SD = 0.96$).

Conspiracy mentality ($\alpha = .84$; $M = 5.71$, $SD = 2.26$) was measured with Bruder et al.' (2013) five-item CMQ (e.g., "[I think that...] ...many very important things happen in the world, which the public is never informed about"), using a response scale from *certainly not* (1) to *certain* (11).

Covariates. Participants' year of birth, gender, highest level of completed education, religiosity (from *not at all religious* (1) to *very religious* (7)), nationality, and political orientation (general, social, and economic; from *extremely liberal* (1) to *extremely conservative* (9)) were measured as covariates.

4.2. Results

Pearson's r bivariate correlation coefficients for the relationships between the main variables can be found in the Supplement (Table S1).

4.2.1. Independent samples t -tests

4.2.1.1. Thinking styles. In support of pre-registered hypothesis 1, an independent samples t -test indicated that the treatment condition significantly increased levels of AOT compared to the control group, $t(460) = 2.56, p = .011, d = 0.25$ (see Fig. 1). Furthermore, to assess our exploratory CRT hypothesis, another independent samples t -test indicated that the treatment condition significantly increased levels of CRT compared to the control group, $t(460) = 3.02, p = .003, d = 0.29$ (see Fig. 1).

4.2.1.2. Conspiracy beliefs. A confirmatory factor analysis revealed that global fit was excellent and significantly better when the GCBS was split into its sub-factors, $\chi^2(66) = 266.53, p < .001$; CFI = 0.95; TLI = 0.93; RMSEA = 0.08, 95 % CI [0.07, 0.09]; SRMR = 0.06, compared to when it was measured as a single factor, $\chi^2(77) = 851.11, p < .001$; CFI = 0.80; TLI = 0.76; RMSEA = 0.15, 95 % CI [0.14, 0.16]; SRMR = 0.09; $\Delta\chi^2(11) = 584.58, p < .001$. Therefore, the GCBS was analysed by distinguishing between its five sub-factors in all subsequent analyses, as pre-registered. As a pre-cursor to our pre-registered indirect path analyses, independent samples t -tests confirmed that the treatment condition significantly reduced overall conspiracy beliefs measured by the GCBS, $t(460) = 2.45, p = .014, d = 0.24$, as well as beliefs in the GCBS sub-scales of malevolent global conspiracies, $t(460) = 2.97, p = .003, d = 0.28$, extra-terrestrial cover-ups, $t(460) = 3.37, p < .001, d = 0.32$, and personal wellbeing, $t(460) = 2.06, p = .040, d = 0.20$ (see Fig. 2). However, the treatment condition did not significantly reduce conspiracy beliefs regarding the control of information, $t(460) = 1.04, p = .299, d = 0.10$, or government malfeasance, $t(460) = 0.82, p = .412, d = 0.08$, compared to the control group.

4.2.1.3. Misinformation susceptibility. As a pre-cursor to our pre-registered indirect path analyses, independent samples t -tests indicated that the treatment condition did not significantly improve the detection of fake news headlines compared to the control group, $t(460) = 1.09, p = .268, d = 0.11$, and its effects on the detection of real news headlines, $t(460) = 1.85, p = .065, d = 0.18$, and veracity discernment, $t(460) = 1.96, p = .051, d = 0.19$, were not significant. Furthermore, the treatment condition (vs. control) had no effect on general distrust, $t(456) = 0.65, p = .514, d = 0.06$.¹

4.2.1.4. Sharing decisions. Analysis of Skewness and Kurtosis indicated that the data for willingness to share fake news headlines was positively skewed, Skewness = 1.28, and the data for willingness to share real news headlines showed negative Kurtosis, Kurtosis = -1.10. As a pre-cursor to our pre-registered indirect path analyses, independent samples t -tests indicated that the treatment condition did not significantly increase intentions to share real news headlines compared to the control group, $t(458) = 0.36, p = .719, d = 0.03$, but it significantly reduced intentions to share fake news headlines, $t(458) = 2.88, p = .004, d = 0.28$, and

significantly improved sharing decisions, $t(458) = 2.71, p = .007, d = 0.26$ (see Fig. 3).²

4.2.2. Indirect path model

To test our pre-registered indirect hypotheses, a bias-corrected bootstrap indirect path model was constructed with a condition coded variable of the treatment group (1) vs. the control group (-1) as the predictor, both AOT and CRT as simultaneous covaried mediators, and veracity discernment, sharing decisions, and all types of conspiracy beliefs as simultaneous outcome variables (see Fig. 4).³ Of the direct effects in this model, the AOT inoculation (vs. control) positively predicted AOT, CRT, and sharing decisions, and negatively predicted belief in malevolent global conspiracies and belief in extra-terrestrial cover-ups. Furthermore, AOT positively predicted veracity discernment and sharing decisions, and negatively predicted belief in malevolent global conspiracies, belief in extra-terrestrial cover-ups, and belief in conspiracies against personal wellbeing. Finally, CRT negatively predicted sharing decisions, and positively predicted belief in conspiracies regarding the control of information and government malfeasance.

The AOT inoculation (vs. control) had positive significant total effects on veracity discernment and sharing decisions, as well as negative total effects on conspiracy beliefs regarding extra-terrestrial cover-ups, malevolent global conspiracies, and personal wellbeing, $\beta = -.10, p = .027$ (see Fig. 4). Furthermore, the AOT inoculation (vs. control) had positive indirect effects on veracity discernment, standardised indirect effect = 0.06, 95 % CI [0.01, 0.12], and sharing decisions, standardised indirect effect = 0.04, 95 % CI [0.01, 0.07], and negative indirect effects on conspiracy beliefs regarding malevolent global conspiracies, standardised indirect effect = -0.05, 95 % CI [-0.08, -0.01], extra-terrestrial cover-ups, standardised indirect effect = -0.03, 95 % CI [-0.06, -0.01], and personal wellbeing, standardised indirect effect = -0.04, 95 % CI [-0.07, -0.01], through AOT. Furthermore, the AOT inoculation (vs. control) had a negative indirect effect on sharing decisions, standardised indirect effect = -0.04, 95 % CI [-0.07, -0.01], and a positive indirect effect on conspiracy beliefs regarding government malfeasance, standardised indirect effect = 0.04, 95 % CI [0.01, 0.08], and the control of information, standardised indirect effect = 0.03, 95 % CI [0.01, 0.05], through CRT.

4.3. Discussion

In Study 1, we found that our AOT inoculation significantly increased AOT, CRT, and sharing decisions compared to the control group, and significantly decreased certain conspiracy beliefs. While it did not directly increase veracity discernment, it did so indirectly through AOT in an indirect path model.

Interestingly, while AOT was positively related to sharing decisions and negatively related to certain conspiracy beliefs, CRT was negatively related to sharing decisions and positively related to conspiracy beliefs regarding government malfeasance and the control of information. These findings suggest that, in contrast to Yelbuz et al.' (2022) meta-analysis, the simple act of cognitive reflection alone is not enough to reduce susceptibility to misinformation (see also Roozenbeek, Maertens, et al., 2022; Večkalov et al., 2024). In fact, our findings suggest that CRT makes individuals *more* likely to believe certain conspiracy narratives and

¹ While our hypotheses detailed our expectations for the detection of and willingness to share fake and real news headlines separately, we altered our main analyses to capture veracity discernment and sharing decisions as more informative measures of misinformation susceptibility. Specifically, analysis of the effects on the detection and willingness to share real vs. fake news separately does not allow for an assessment of improved detection and reduced willingness to share fake news relative to real news. Total effects for both analyses are reported here, and results for indirect path models on these variables separately can be found in the Supplement (Figures S1 and S2).

² A robustness check indicated that scores on the CMQ were significantly lower in the treatment condition, $M = 5.40, SD = 2.17$, compared to the control group, $M = 5.89, SD = 2.29, t(451) = 2.25, p = .025, d = 0.11$, suggesting that our moderator of general conspiracy beliefs was contaminated by the experimental manipulation.

³ The covariates of conspiracy mentality and socio-demographics did not correlate with both the predictors and outcome variables on each respective path, and therefore we did not control for the shared variance between these measures.

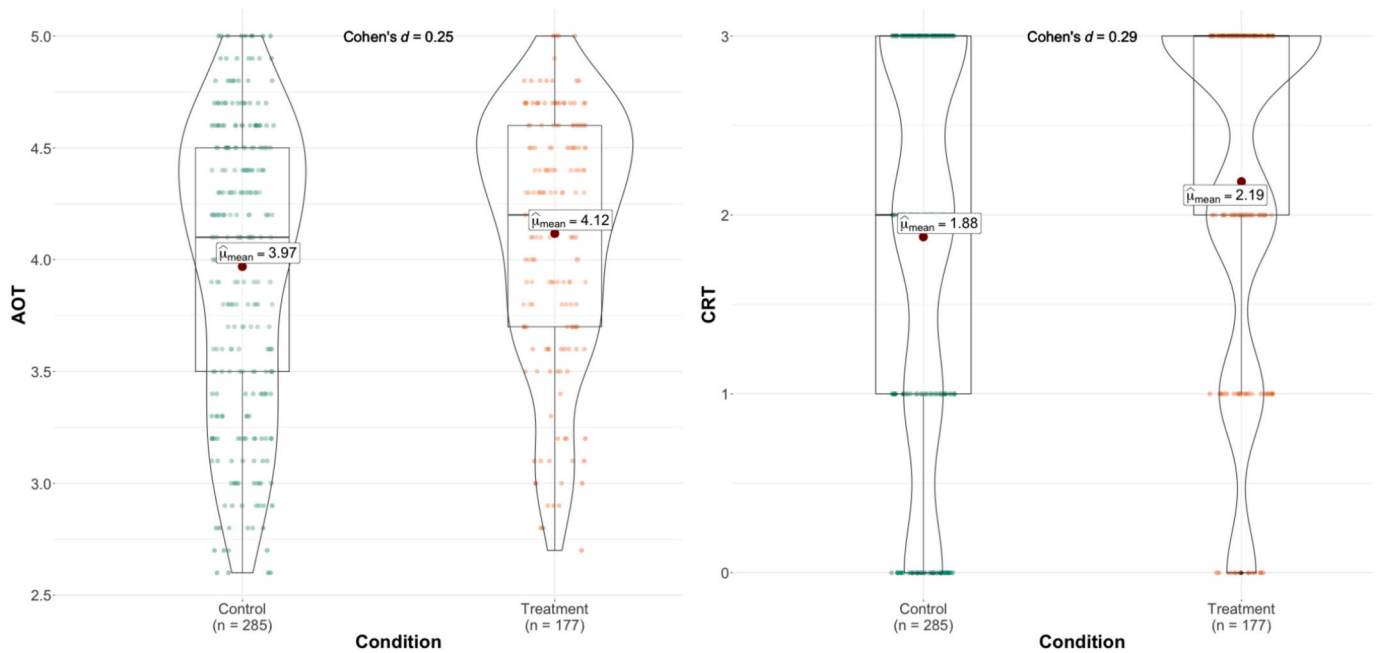


Fig. 1. Violin Plots for the Mean Differences in AOT (Left) and CRT (Right) between the Treatment and Control Conditions (Study 1).

share misinformation when controlling for the shared variance with AOT. Therefore, the additional components to AOT are likely to confirm it as a promising thinking style to harness in the fight against misinformation. We posited that these improvements to our experimental manipulation should yield more reliable changes in participants' psychological states.

5. Study 2

In Study 1, the AOT inoculation improved sharing decisions and reduced certain conspiracy beliefs. Furthermore, the treatment (vs. control) indirectly improved veracity discernment. As such, our second study provided an opportunity to further understand the exact components that are most effective at making AOT inoculation a promising strategy to reduce misinformation susceptibility. In addition to our improved experimental manipulation, we also opted to include an active control group to extend on our findings with the passive control group used in Study 1. We suggest that the neutral word sorting task used by Maertens et al. (2020, 2025) is a promising candidate for an active control condition. This task provides a better match for the experimental manipulation by matching its structure and duration, ensuring that participants in both conditions are similarly attentive and cognitively activated. At the same time, it is content-neutral, without any references to misinformation, epistemic reasoning, or social norms. This allows us to rule out potential confounds such as task-induced priming, motivational differences, or general engagement effects.

In Study 2, we used a similar design to Study 1, this time on a sample of Prolific workers, slightly extending the inoculation message to explain another proposed component of AOT: intolerance of uncertainty. Furthermore, alongside measuring AOT and CRT after the administration of the AOT inoculation, we also included other measures of the proposed components of AOT: an intellectual humility (IH) scale, an intolerance of uncertainty scale (IUS), an illusion of explanatory depth (IOED) task, a need for closure (NFC) scale, and an ideological bias awareness (IBA) scale. This would allow us to determine which precise cognitive mechanism(s) were most important in our intervention process. Once again, we measured participants' misinformation susceptibility and conspiracy beliefs to replicate the effects of Study 1. While in Study 1 the significant reduction in conspiracy beliefs was only found for

certain types of conspiracy narratives, we chose to include a 5-item general measure of conspiracy beliefs here to reduce the time taken to complete the survey and to test whether our AOT inoculation reduces the general tendency to engage in conspiracy theorising. In line with our findings from Study 1, we hypothesised that the AOT inoculation would significantly increase AOT compared to the control group, and that it would indirectly reduce conspiracy beliefs and increase the detection of real news headlines. We pre-registered exploratory hypotheses regarding the other outcome measures. The hypotheses, design, and analyses were fully pre-registered and can be found at https://aspredict.org/BF5_B47.

5.1. Methods

5.1.1. Participants

According to Fritz and MacKinnon (2007), a sample size of 462 is required to detect a bias-corrected indirect effect with small coefficients on all paths with a power of .80. Therefore, we aimed to collect a minimum sample of $N = 462$ ($N = 231$ per condition).

A total of 571 responses of US residents were collected from Prolific. Once participants were removed for failing attention checks, the final sample was 464 (223 women, 230 men, $M_{age} = 40.85$; $SD_{age} = 13.80$). Sensitivity power analysis using the "statsmodels" module in Python indicated that a power of .90 was achieved to detect a minimum effect size of $d = 0.30$, and a power of .80 was achieved to detect a minimum effect size of $d = 0.26$. Furthermore, according to analyses by Fritz and MacKinnon (2007), we achieved enough power to conduct an indirect path analysis with small coefficients on both the a and b paths.

5.1.2. Design and procedure

After reading the information sheet and providing their consent, participants were randomly allocated to the treatment or control condition. Afterwards, they were presented with the measures of AOT, CRT, numeracy ability, IH, IUS, IOED, NFC, and IBA in randomised order, and were then asked to complete the measures of conspiracy beliefs and misinformation susceptibility in randomised order. Finally, participants filled out the covariate information before being fully debriefed.

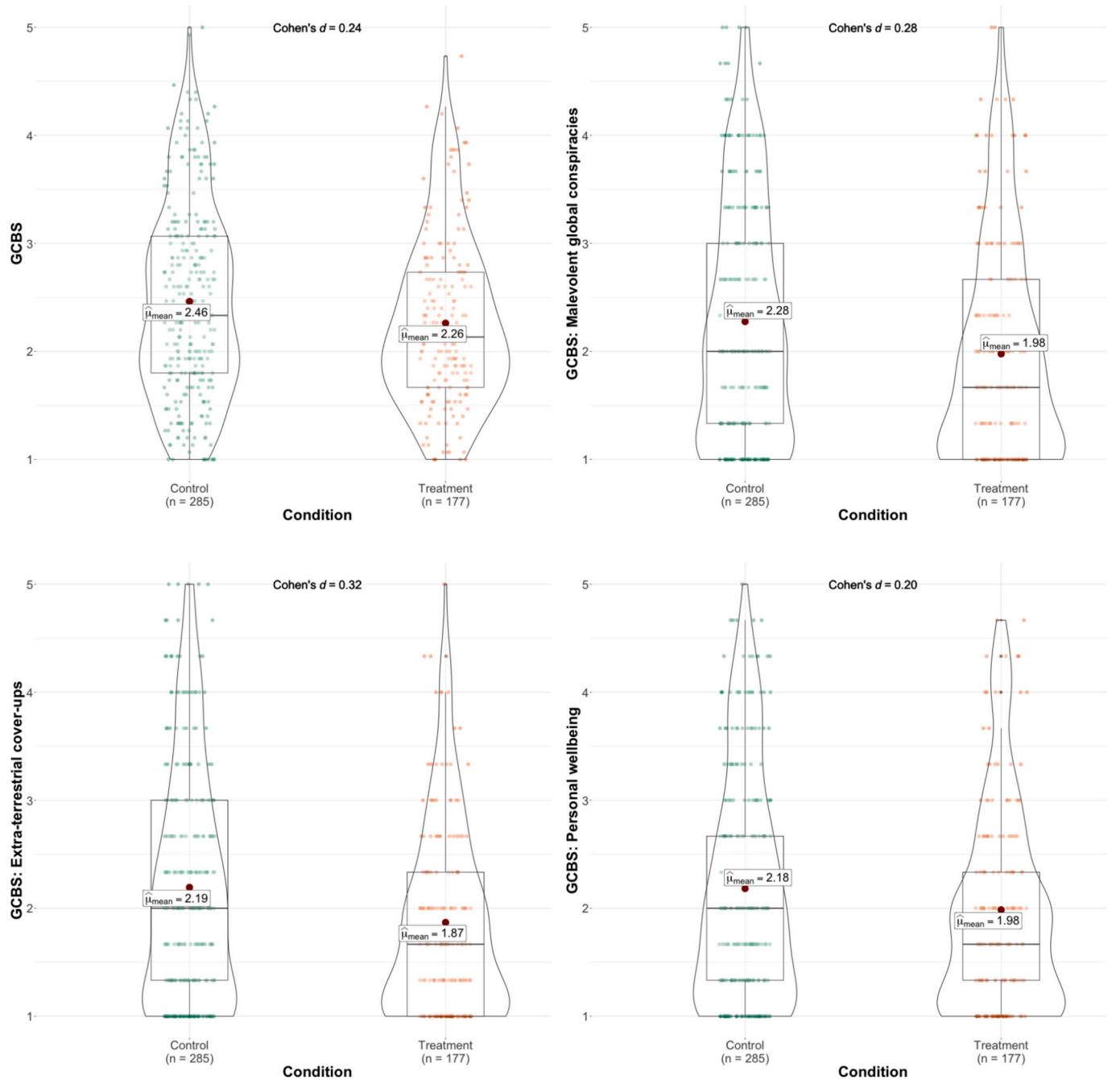


Fig. 2. Violin Plots for the Mean Differences in the Overall GCBS (Top Left), and GCBS Sub-Scales of Malevolent Global Conspiracies (Top Right), Extra-Terrestrial Cover-Ups (Bottom Left), and Personal Wellbeing (Bottom Right) between the Treatment and Control Conditions (Study 1).

5.2. Materials

The norm-enhanced inoculation against failure to engage in AOT was almost identical to the one administered in Study 1, with two minor alterations. First, an additional component of AOT was added to the description to allude to an intolerance of uncertainty (“A need to find answers when faced with ambiguity”). Second, the final phrase was altered to read “don’t believe everything you *initially* think!”. In the control condition, participants were asked to complete a neutral word sorting task (see Supplement, p. 25). This control condition was adopted from Maertens et al. (2020, 2025).

5.3. Measures

Actively open-minded thinking ($\alpha = .82$; $M = 4.05$, $SD = 0.58$) was measured with Baron et al.’ (2022) 11-item AOT scale (e.g., “Willingness to be convinced by opposing arguments is a sign of good character”), using a response scale from *completely disagree* (1) to *completely agree* (5).

Cognitive reflection ($\alpha = .63$; $M = 2.07$, $SD = 1.04$) was measured as in Study 1. However, this time, responses were collected using a multiple choice scale, and participants were told that they had a maximum of 3 minutes to complete the questions, after which the page would automatically progress.

Numeracy ability ($\alpha = .44$; $M = 2.18$, $SD = 0.88$) was measured with three items from the Berlin Numeracy Test (Cokely et al., 2012; e.g., “In the BIG BUCKS LOTTERY, the chance of winning a \$10 prize is

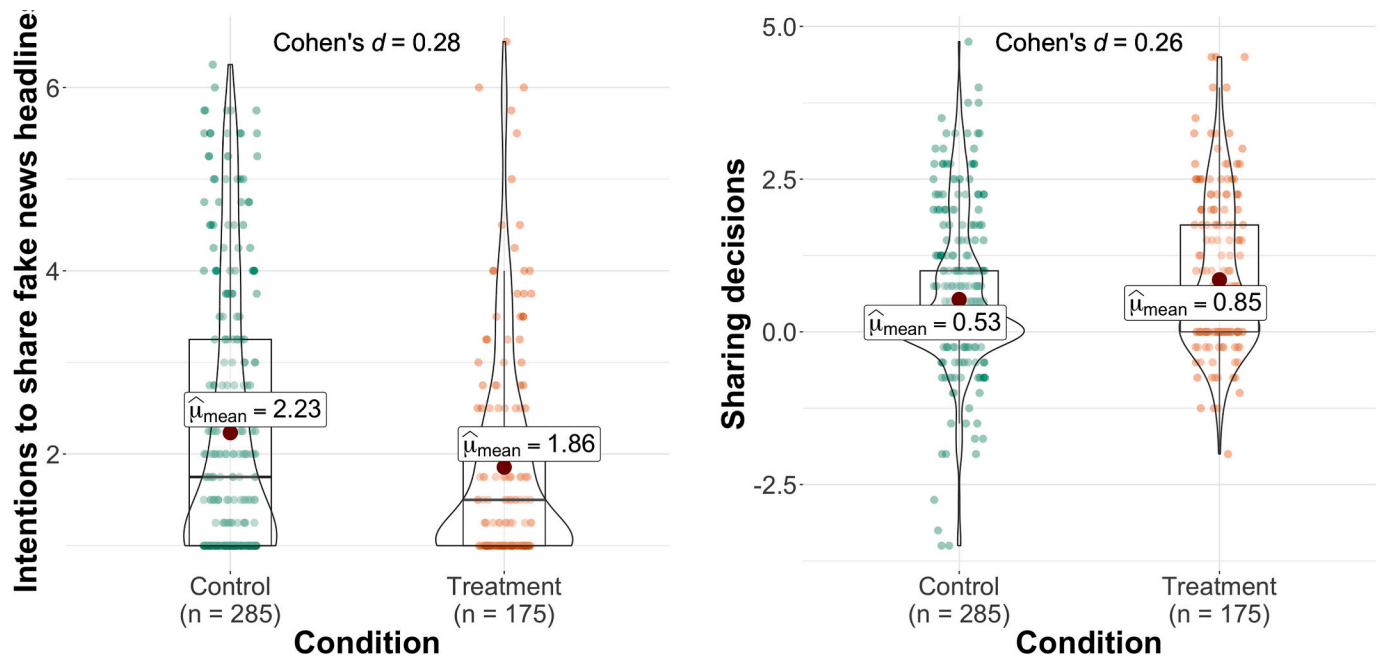


Fig. 3. Violin Plots for the Mean Differences in Intentions to Share Fake News Headlines (Left) and Sharing Decisions (Right) between the Treatment and Control Conditions (Study 1).

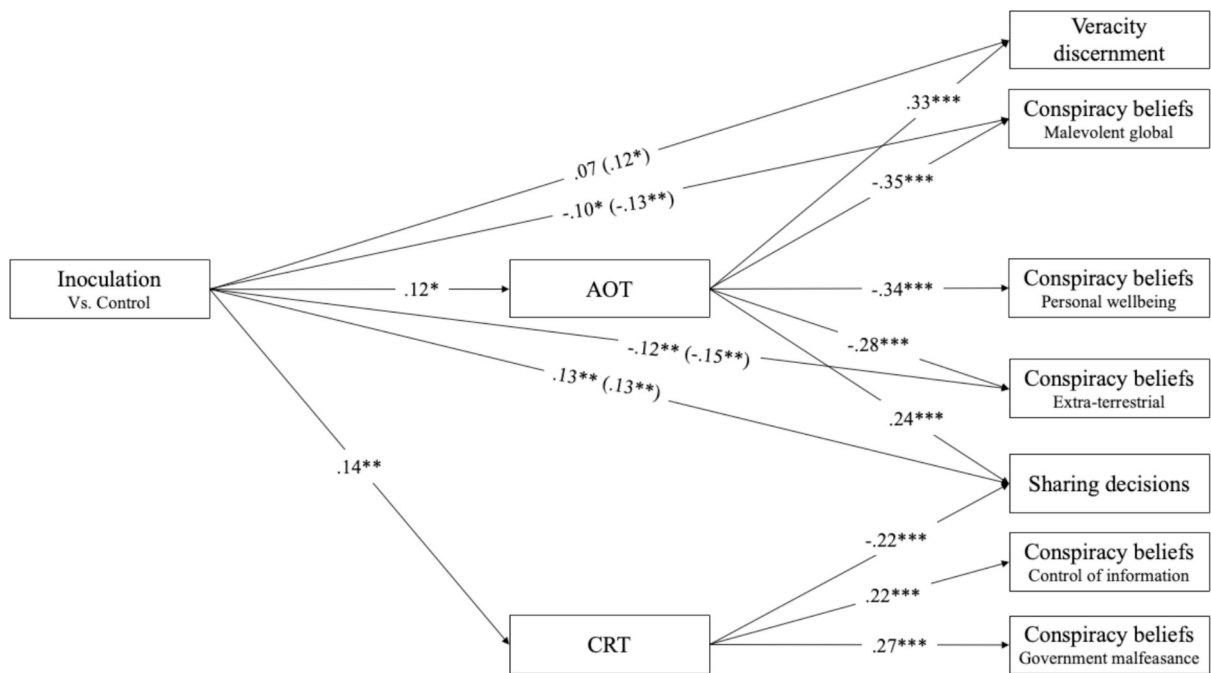


Fig. 4. Indirect Path Model Detailing the Relationships Between the AOT Inoculation (vs. Control) and Veracity Discernment, Sharing Decisions, and all types of Conspiracy Beliefs through both AOT and CRT (Study 1). Note. * $p < .05$; ** $p < .01$; *** $p < .001$. All paths are standardised coefficients. Total effects are displayed in parentheses.

1%. What is your best guess about how many people would win a \$10 prize if 1000 people each buy a single ticket to BIG BUCKS?). Correct (1) and incorrect (0) answers were summed so that higher scores corresponded to great numeracy ability.

Intellectual humility ($\alpha = .81$; $M = 4.04$, $SD = 0.57$) was measured with Leary et al.' (2017) six-item intellectual humility scale (e.g., "I accept that my beliefs and attitudes may be wrong") using a response scale from *strongly disagree* (1) to *strongly agree* (5).

Intolerance of uncertainty ($\alpha = .89$; $M = 3.14$, $SD = 0.77$) was

measured with Carleton et al.' (2007) IUS (e.g., "Unforeseen events upset me greatly"), using a response scale from *strongly disagree* (1) to *strongly agree* (5).

The **illusion of explanatory depth** ($\alpha = .90$; $M = 0.15$, $SD = 0.71$) was measured using an adapted version of Rozenblit and Keil's (2002) paradigm (see also Fernbach et al., 2013), wherein participants are first asked to estimate their understanding of a given topic (in our case, the impact of raising the minimum wage) using a scale from *very vague understanding* (1) to *very thorough understanding* (7). Next, participants

were given a maximum of one minute to describe all the details they know about the impact of raising the minimum wage in the text box provided before the page would automatically progress. Finally, participants were asked again to estimate their understanding of raising the minimum wage using the same response scale as the first time. The second estimate was subtracted from the first estimate so that positive scores indicated greater IOED and negative scores indicated weaker IOED. Analysis of Skewness and Kurtosis indicated that IOED scores showed extreme positive Kurtosis, Kurtosis = 8.20.

Need for closure was measured with Roets and Van Hiel's (2011) 15-item NFC scale (e.g., "I don't like situations that are uncertain"), using a response scale from *completely disagree* (1) to *completely agree* (5). This scale consists of five sub-scales: ambiguity ($\alpha = .65$; $M = 3.50$, $SD = 0.84$), closed-mindedness ($\alpha = .54$; $M = 2.57$, $SD = 0.84$), order ($\alpha = .84$;

$M = 3.85$, $SD = 0.83$), predictability ($\alpha = .77$; $M = 3.56$, $SD = 0.90$), and decisiveness ($\alpha = .68$; $M = 3.45$, $SD = 0.84$).

Ideological bias awareness is a metacognitive belief structure about the extent of one's own ideological bias, including three subscales which capture if individuals a) recognise that they hold and apply bias (*bias recognition*, $\alpha = .81$; $M = 3.30$, $SD = 0.84$), b) are concerned and worried about it (*bias concernedness*, $\alpha = .87$; $M = 2.83$, $SD = 0.97$) and c) are motivated to control their bias expression (*bias self-control*, $\alpha = .79$; $M = 3.67$, $SD = 0.77$). IBA was measured with a scale currently under validation (see Ziemer et al., 2025). The 15-item scale (e.g., "I am biased when evaluating scientific findings about political topics") was measured using a response scale from *completely disagree* (1) to *completely agree* (5).

Misinformation susceptibility was measured as in Study 1,

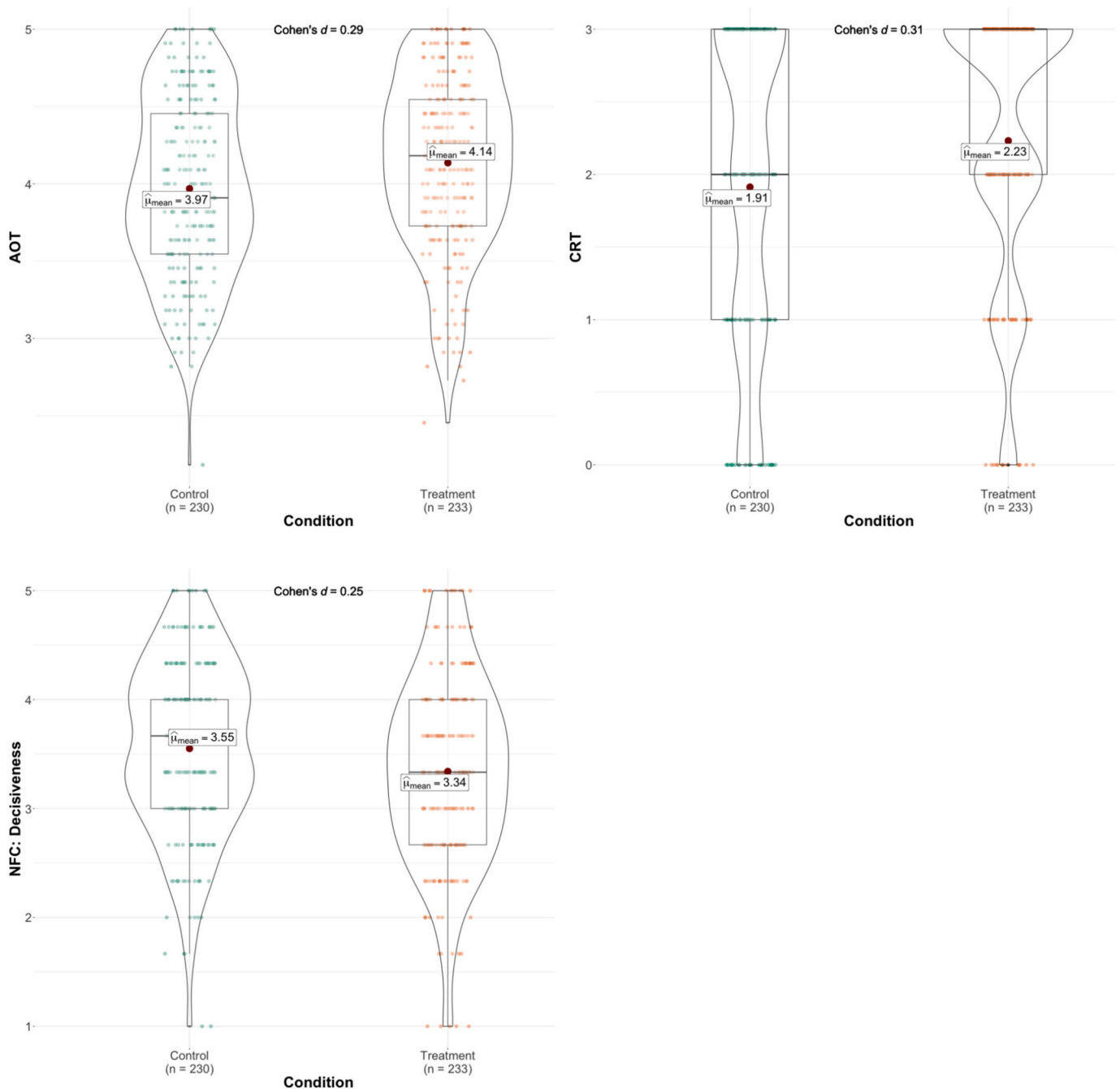


Fig. 5. Violin Plots for the Mean Differences in AOT (Top Left), CRT (Top Right), and NFC Decisiveness (Bottom Left) between the Treatment and Control Conditions (Study 2).

capturing participants' detection of four fake ($\alpha = .58$; $M = 2.89$, $SD = 1.17$) and four real ($\alpha = .52$; $M = 3.07$, $SD = 1.07$) news headlines, which were used to calculate veracity discernment ($M = 5.96$, $SD = 1.81$). Analysis of Skewness and Kurtosis indicated that the data for the detection of real news headlines was negatively skewed, Skewness = -1.09 .

Conspiracy beliefs ($\alpha = .86$; $M = 2.90$, $SD = 1.04$) were measured with Kay and Slovic's (2023) five-item Generic Conspiracist Beliefs (GCB-5) scale (e.g., "Certain significant events have been the result of the activity of a small group who secretly manipulate world events"), using a response scale from *definitely not true* (1) to *definitely true* (5).

Covariates were measured as in Study 1.

5.4. Results

Pearson's r Bivariate correlation coefficients for the relationships between the main variables can be found in the Supplement (Table S10).

5.4.1. Independent samples t-tests

5.4.1.1. Thinking styles. As pre-registered, an independent samples t-test indicated that the treatment condition significantly increased levels of AOT compared to the control group, $t(461) = 3.13$, $p = .002$, $d = 0.29$ (see Fig. 5). Furthermore, to assess our exploratory pre-registered analyses, independent samples t-tests indicated that the treatment condition

significantly increased CRT compared to the control group, $t(461) = 3.33$, $p < .001$, $d = 0.31$, and significantly reduced NFC decisiveness compared to the control group, $t(461) = 2.71$, $p = .007$, $d = -0.25$ (see Fig. 5). The treatment condition did not significantly alter numeracy ability, $t(461) = 1.13$, $p = .258$, $d = 0.11$, IH, $t(461) = 0.05$, $p = .959$, $d = 0.00$, IUS, $t(461) = 0.39$, $d = 0.04$, $p = .698$, IOED, $t(461) = 0.94$, $p = .348$, $d = 0.09$, NFC ambiguity, $t(461) = 0.96$, $p = .336$, $d = 0.09$, NFC closure, $t(461) = 0.11$, $p = .910$, $d = 0.01$, NFC order, $t(461) = 0.37$, $p = .708$, $d = 0.03$, NFC predictability, $t(461) = 0.09$, $p = .931$, $d = 0.01$, IBA recognition, $t(461) = 1.49$, $p = .136$, $d = 0.14$, IBA concernedness, $t(461) = 0.42$, $p = .672$, $d = 0.04$, or IBA self-control, $t(461) = 1.48$, $p = .141$, $d = 0.14$, compared to the control group.

5.4.1.2. Conspiracy beliefs. As a pre-cursor to our pre-registered indirect path analysis, an independent samples t-test indicated that the treatment condition did not significantly alter conspiracy beliefs compared to the control group, $t(461) = 1.03$, $p = .305$, $d = 0.10$.

5.4.1.3. Misinformation susceptibility. As a pre-cursor to our pre-registered indirect path analysis, independent samples t-tests indicated that the treatment condition significantly improved both the detection of fake news headlines, $t(461) = 4.01$, $p < .001$, $d = 0.37$, and veracity discernment, $t(461) = 3.40$, $p < .001$, $d = 0.32$, and general distrust, $t(461) = 2.41$, $p = .017$, $d = 0.22$, compared to the control group (see Fig. 6). Furthermore, the treatment condition did not significantly alter

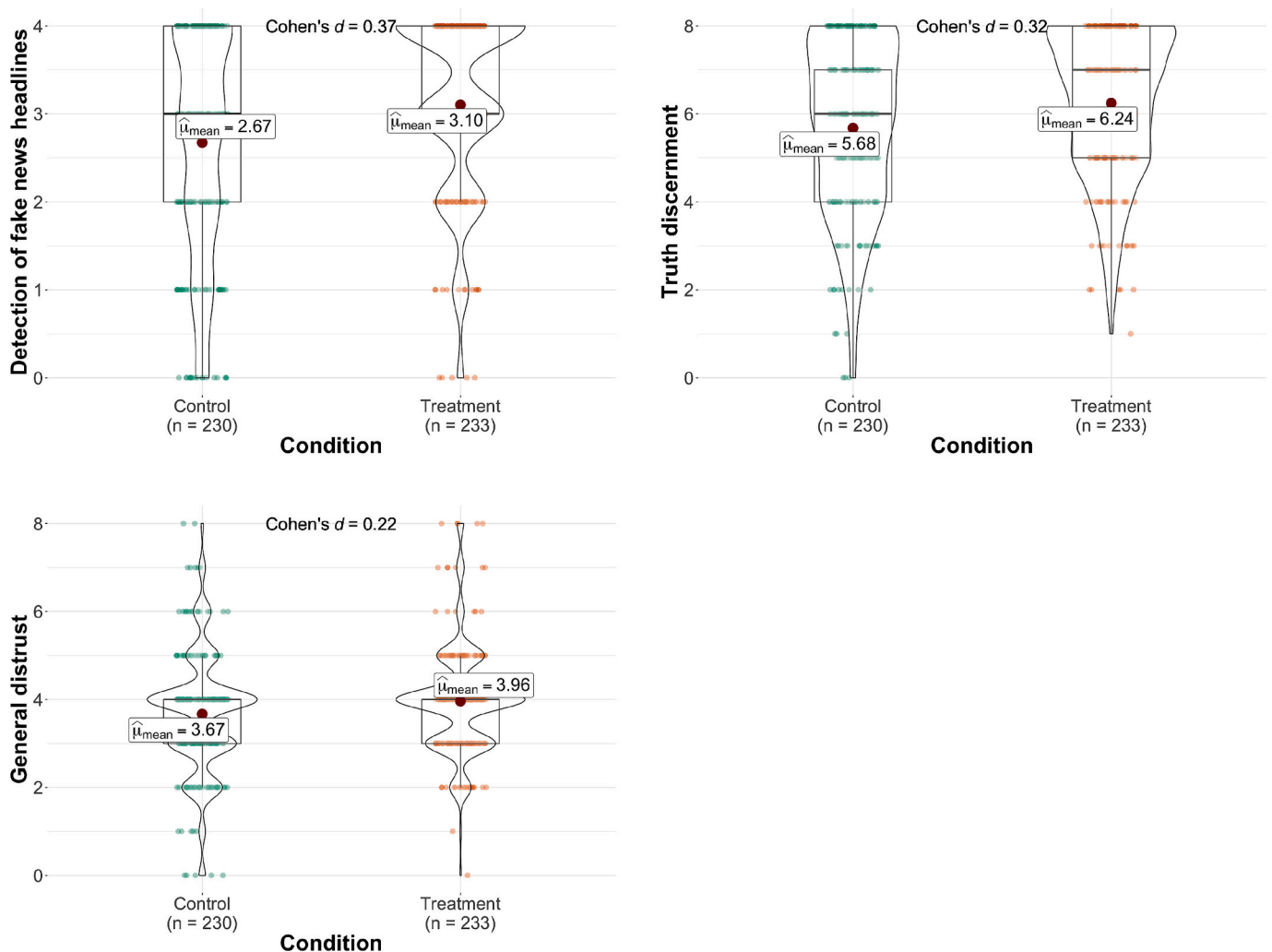


Fig. 6. Violin Plots for the Mean Differences in the Detection of Fake News Headlines (Top Left), Veracity Discernment (Top Right), and General Distrust (Bottom Left) between the Treatment and Control Conditions (Study 2).

the detection of real news headlines compared to the control group, $t(461) = 1.39$, $p = .166$, $d = 0.13$.

5.4.2. Indirect path model

Next, we tested our pre-registered indirect path analysis by constructing an indirect path model, with a condition coded variable of the experimental effect (1 = treatment condition, -1 = control condition) as the predictor, AOT, CRT, numeracy ability, IH, IUS, IOED, the NFC subscales, and IBA subs-scales as simultaneous mediators, and both veracity discernment and conspiracy beliefs as simultaneous outcome variables. In this model, the AOT inoculation (vs. control) positively predicted AOT, CRT, and veracity discernment, and negatively predicted NFC decisiveness (see Fig. 7). Furthermore, conspiracy beliefs were negatively predicted by AOT and numeracy ability, and positively predicted by IH. Finally, veracity discernment was positively predicted by AOT, numeracy ability, NFC predictability, and IBA concernedness, and negatively predicted by NFC order.

While the AOT inoculation (vs. control) also had a positive total effect on veracity discernment (see Fig. 7), it had no total effect on conspiracy beliefs, $\beta = -.05$, $p = .311$. Furthermore, the AOT inoculation had a positive indirect effect on veracity discernment, standardised indirect effect = 0.06, 95 % CI [0.03, 0.17], and a negative indirect effect on conspiracy beliefs, standardised indirect effect = -0.04, 95 % CI [-0.07, -0.01], through AOT.⁴

5.5. Discussion

In Study 2, we sought to replicate and extend the results of Study 1. Once again, the AOT inoculation significantly increased AOT compared to the control group. Unlike Study 1, the AOT inoculation significantly increased veracity discernment, but did not directly reduce conspiracy beliefs. However, in the indirect path model, the AOT inoculation (vs. control) indirectly reduced conspiracy beliefs and increased veracity discernment. While the AOT inoculation (vs. control) also increased CRT and reduced NFC decisiveness, neither of these variables were related to veracity discernment or conspiracy beliefs. Furthermore, veracity discernment was positively predicted by NFC predictability, IBA concernedness, and numeracy ability, and negatively predicted by NFC order. Conspiracy beliefs were also positively predicted by IH and negatively predicted by numeracy ability. However, none of these variables were altered by the AOT inoculation.

The findings from Study 2 largely replicated those from Study 1. Importantly, unlike AOT, CRT was only increased by the AOT inoculation (vs. control) but did not subsequently predict any of the outcome variables. This suggests that while in Study 1 there were indirect effects for both AOT and CRT, these effects only appear to be consistent for AOT. In fact, AOT was the only mediator between the AOT inoculation (vs. control) and both veracity discernment and conspiracy beliefs, suggesting that there is something unique about the multifaceted nature of AOT that appears to make it greater than the sum of its component parts.

6. General discussion

Recent work has demonstrated how AOT—a cognitive style incorporating a myriad of effective thinking strategies, including a willingness to change one's views in the face of refuting evidence and consistent reflection on one's own attitudes—is a better predictor of low misinformation susceptibility than reflective thinking alone (see Roozenbeek, Maertens, et al., 2022). Therefore, we set out to develop a logic-based inoculation that could be used to improve participants'

willingness to engage in AOT, and in turn, reduce their misinformation susceptibility. In both Studies 1 and 2, the AOT inoculation significantly increased AOT scores relative to the control groups, and in turn, reduced conspiracy beliefs and improved veracity discernment of news headlines. Furthermore, in Study 1, the AOT inoculation (vs. control) directly decreased certain conspiracy beliefs and improved sharing decisions. When analysed separately, the AOT inoculation significantly reduced willingness to share both fake and real news headlines. This suggests that while our intervention is likely to improve the overall veracity of information that people share online, it also makes people less likely to share content in general. This likely reflects an increase in participants' hesitation to spread information to others before collecting more information about the topic. In Study 2, the AOT inoculation also directly increased veracity discernment. Therefore, our findings support the use of logic-based inoculation to reduce misinformation susceptibility by improving AOT.

In both studies, the AOT inoculation not only increased AOT, but it also increased CRT. In Study 2, the AOT inoculation also reduced NFC decisiveness and increased general distrust compared to the control group. These findings suggest that the AOT inoculation may also have unintended effects in other domains. However, while CRT was positively related to conspiracy beliefs regarding the control of information and government malfeasance in Study 1, both CRT and NFC decisiveness were unrelated to conspiracy beliefs or veracity discernment in Study 2. Furthermore, considering the increase in veracity discernment alongside the small increase in general distrust in Study 2, this unintended effect may reflect reduced gullibility through the motivation to remain vigilant of those seeking to exploit our tendency to not engage in AOT more so than overall distrust of all information encountered after the intervention. Nevertheless, other work has raised concerns over inoculation potentially increasing general distrust or scepticism rather than veracity discernment (e.g., Modirrousta-Galian & Higham, 2023). Thus, more research is required to further understand these apparent inconsistent unintended effects.

6.1. Theoretical implications

Although the AOT inoculation consistently improved overall veracity discernment across both studies, there was an interesting asymmetry when separating out these component effects: in Study 1, the intervention had a stronger effect on improving the detection of real news, whereas in Study 2, it had a stronger effect on improving the detection of fake news. This divergence highlights an important nuance in interpreting the veracity discernment score, which is typically treated as a composite measure reflecting balanced sensitivity to both fake and real content (see Maertens et al., 2023). However, use of this composite score alone without investigating effects on the detection of fake and real news separately may obscure differential pathways through which interventions exert their influence—either by promoting healthy scepticism (reducing belief in fake news) and/or by enhancing confidence in legitimate information (increasing belief in real news; see Acerbi et al., 2022). This pattern suggests that while AOT improves discernment overall, it may do so via different mechanisms depending on contextual factors such as individual differences. Future work should consider analysing and reporting these components both together and separately to better understand how interventions like ours shift cognitive engagement with true versus false content.

Early evidence suggested that simply priming analytic or reflective thinking can reduce conspiracy beliefs (Swami et al., 2014), and more recent similar interventions have implemented *accuracy nudges*, which remind participants to remain accurate to reduce intentions to share misinformation online (see Martel et al., 2024). Despite meta-analytic correlational evidence supporting the link between low reflective thinking and conspiracy beliefs (Yelbuz et al., 2022), experimental replications have provided null results (Večkalov et al., 2024). Instead, Roozenbeek, Maertens, et al. (2022) showed that low AOT was a much

⁴ There were no covariates that correlated with both the predictors and outcome variables on each respective path, and therefore we did not control for the shared variance with any of the socio-demographic measures.

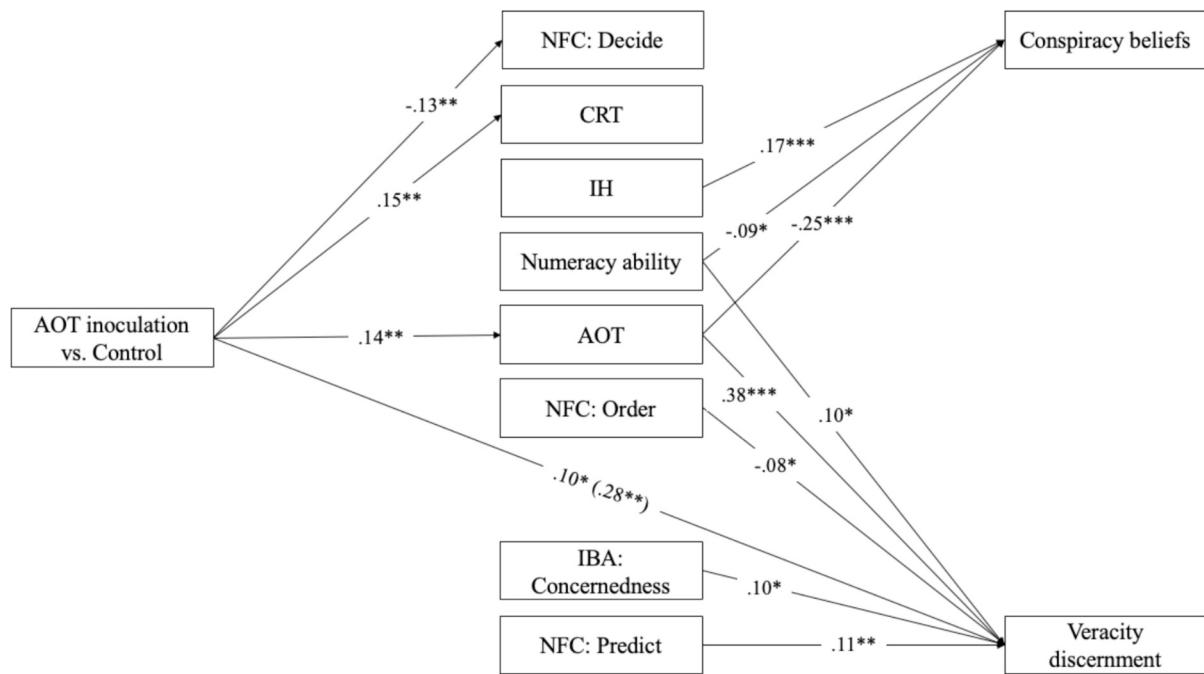


Fig. 7. Segment of the Indirect Path Model for the Effect of the AOT Inoculation (vs. Control) on all Outcome Variables through AOT.

better predictor of intentions to share misinformation and misinformation susceptibility than reflective thinking alone. Our findings suggest that in an intervention setting, this also appears to be the case. Specifically, although our AOT inoculation increased AOT and CRT in both studies, only the improvements in AOT had consistent positive implications for improved veracity discernment and sharing decisions, as well as reduced conspiracy beliefs.

Theoretically, AOT incorporates multiple thinking strategies that make individuals more willing to change their attitudes in the face of refuting evidence and effectively appraise when it is appropriate to search for more information to inform these attitudes (see Baron, 2019). More specifically, Baron (2019) argues that the implementation of AOT can become a cultural norm, involving such components as IH, low IUS, low IOED, low NFC, and low myside bias. However, our findings suggest that when taken together, AOT is more than the sum of its parts. That is, while certain components measured in Study 2 predicted veracity discernment and conspiracy beliefs in different ways, it was the AOT measure specifically that acted as a consistent mediator between the AOT inoculation and our relevant outcome variables. This suggests that when these component thinking styles are combined into an effective thinking strategy—that is, AOT—they form a thinking style that takes on a new meaning, improving one's ability to detect misinformation and conspiracy theories over and above the effects of these individual components measured together. In contrast, when analysed alongside AOT, certain component variables such as CRT (Study 1) and IH (Study 2) were positively correlated with misinformation susceptibility.

Considering the apparent efficacy of this intervention at reducing misinformation susceptibility in both general and conspiracist domains, we suggest that educating individuals on how to improve their AOT with the use of this inoculation framework is likely to protect people against many different forms of epistemically unwarranted beliefs. For example, AOT, conspiracy beliefs, pseudoscientific beliefs, belief in the paranormal, magical thinking, and belief in pseudo-profound bullshit are all strongly correlated with one another (e.g., Čavojská et al., 2019; Huete-Pérez et al., 2022; Lobato et al., 2014; Rizeq et al., 2020). This suggests that teaching people how to think with the use of a similar paradigm to our AOT inoculation may provide blanket protection against these epistemically unwarranted beliefs, reducing the need to tailor interventions to specific contexts and belief types.

6.2. Limitations and future recommendations

In an experimental context, our AOT inoculation was relatively easy to implement despite its arguably lengthy and verbose content. However, in the real world, we can expect people to be less motivated or engaged with our message. Therefore, future research should also focus on scaling our intervention to more efficient and easy-to-administer formats. Examples include infographics that clearly layout the signs and pitfalls of not engaging AOT in a visual format, short-form video content that can be posted on social media, and educational modules that can be taught at school for students to internalise the information more effectively over time.

In Study 1, we included a passive control condition, and in Study 2, we used a minimal control condition requiring participants to complete a word sorting task adopted from Maertens et al. (2020, 2025). One limitation worth noting is that our control groups did not include any mention of misinformation. This raises the possibility that participants in the treatment group may have been more likely to infer the study's purpose or adjust their responses due to heightened salience of misinformation—a potential demand characteristic. In such cases, observed effects may reflect increased attentiveness to misinformation-related content or perceived experimenter expectations, rather than genuine cognitive or attitudinal change driven by the intervention itself. Although the consistency of our AOT effects across both studies (difference in effect size = 0.04) provides some reassurance, and prior research has found limited evidence that demand characteristics meaningfully bias outcomes in survey experiments (Au et al., 2020; Mummolo & Peterson, 2018), this issue remains an open question. Future research would benefit from stronger systematic variation of the content and framing of control conditions to directly examine the potential influence of demand effects on inoculation interventions.

7. Conclusion

The current article presents the first attempt to use logic-based inoculation to improve AOT, and in turn, reduce misinformation susceptibility. In both studies, AOT was consistently improved, which, in turn, reduced misinformation susceptibility, conspiracy beliefs (Studies 1 & 2), and improved sharing decisions (Study 1). Furthermore, these

benefits of improving AOT occurred even when controlling for the apparent component parts of this thinking style. We hope these findings can pave the way for more scalable interventions that the public can be repeatedly exposed to and internalise this thinking strategy that may protect against a myriad of epistemically unwarranted beliefs.

8. Open practices

All studies, measures, manipulations, and data/participant exclusions are reported in this manuscript or the Supplement. All data, R code, and study materials can be found at: https://osf.io/md2ux/?view_only=fa1611c9844e4948b93892e37a03a21c. Full pre-registrations of the study designs, materials, analyses, and sample sizes can be found at https://aspredicted.org/772_BXC for Study 1, and https://aspredicted.org/BF5_B47 for Study 2.

CRedit authorship contribution statement

Mikey Biddlestone: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Carolin-Theresa Ziemer:** Writing – review & editing, Project administration, Methodology, Formal analysis, Conceptualization. **Rakoen Maertens:** Writing – review & editing, Conceptualization. **Jon Roozenbeek:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Sander van der Linden:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

All studies, measures, manipulations, and data/participant exclusions are reported in this manuscript or the Supplement. This research was supported by the Infodemic Grant from the UK Cabinet Office; IRIS Infodemic, Grant/Award Number: #SCH-00001-3391.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jesp.2025.104818>.

Data availability

All data, R code, and study materials can be found at: https://osf.io/md2ux/?view_only=fa1611c9844e4948b93892e37a03a21c

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